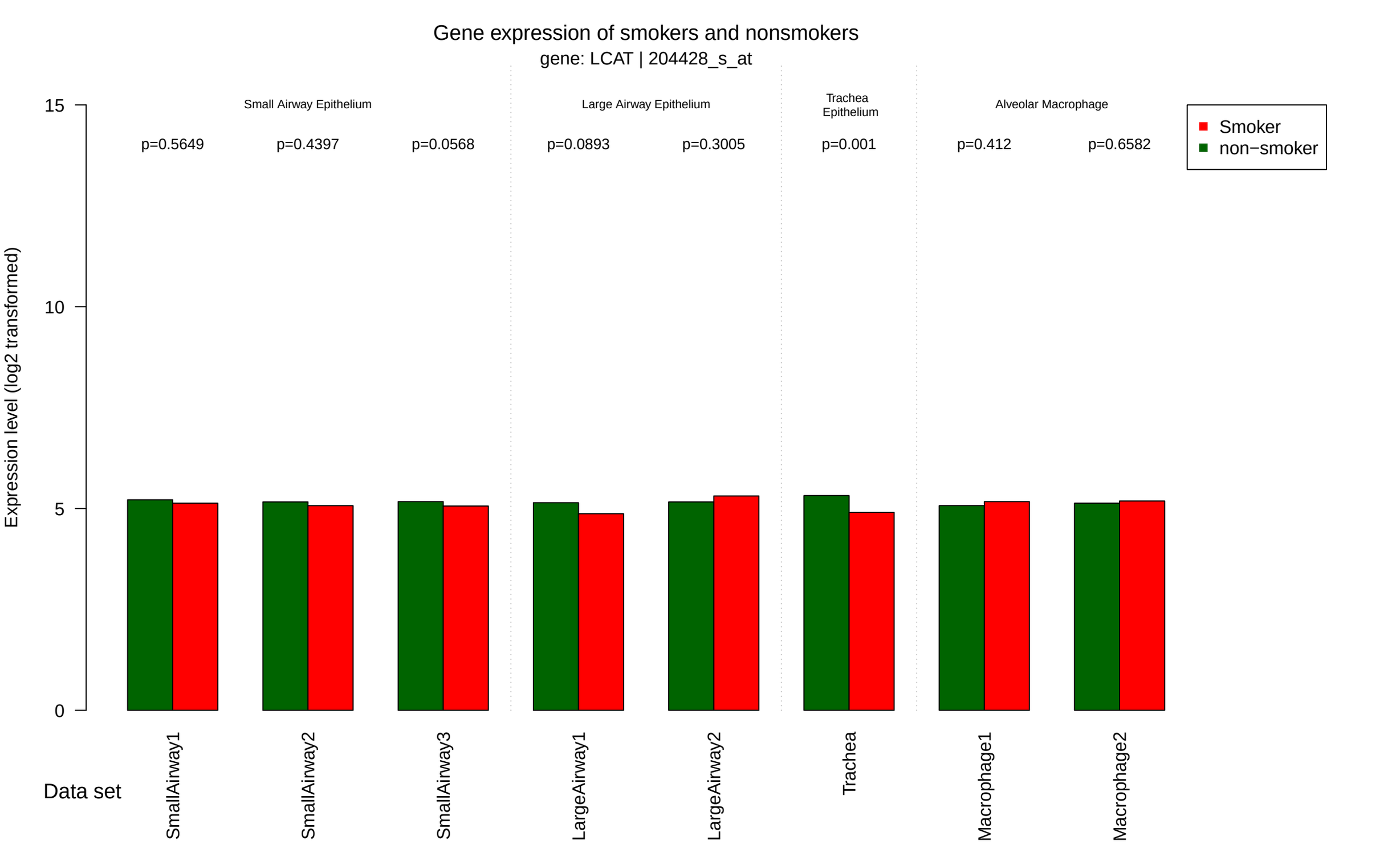
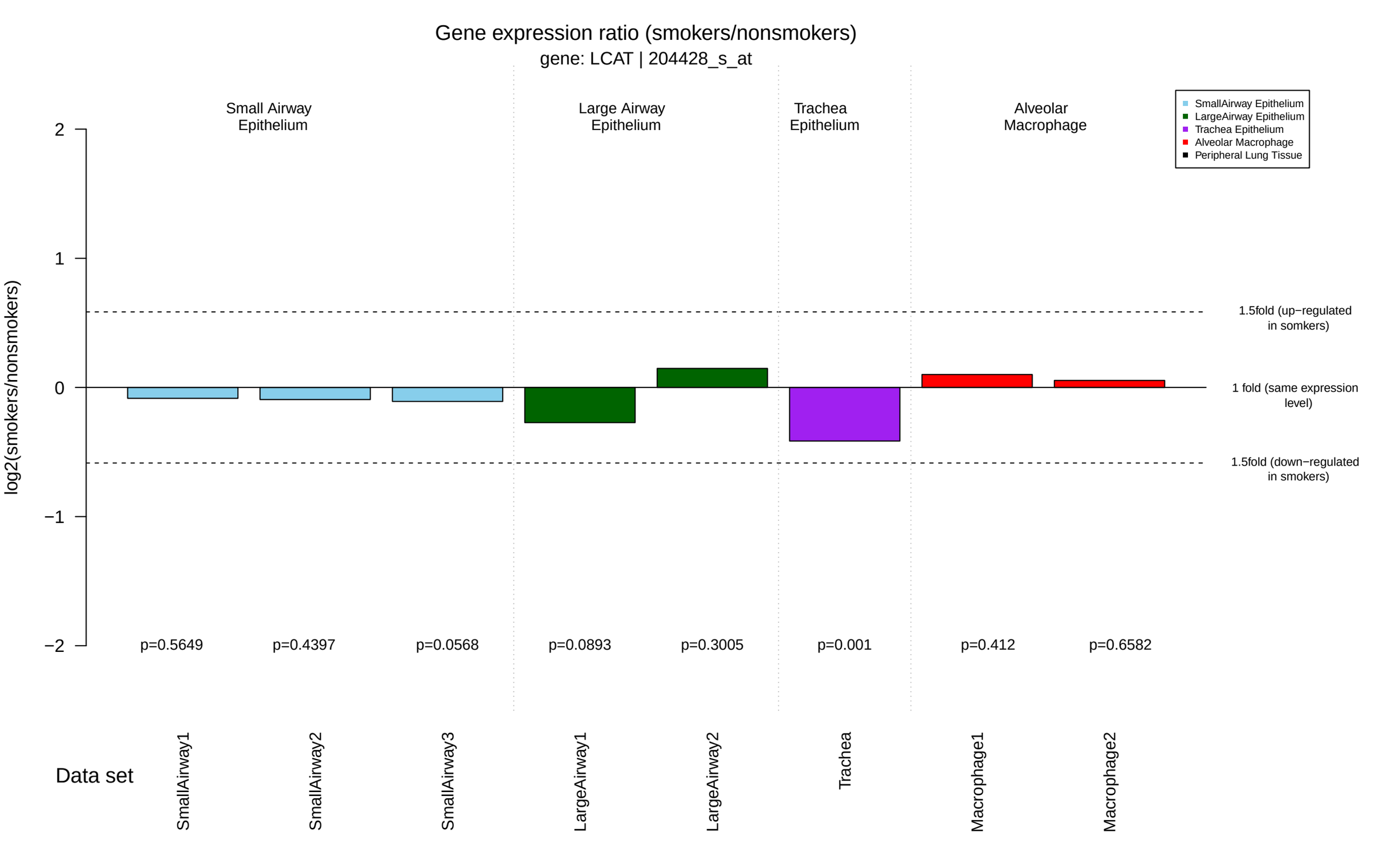
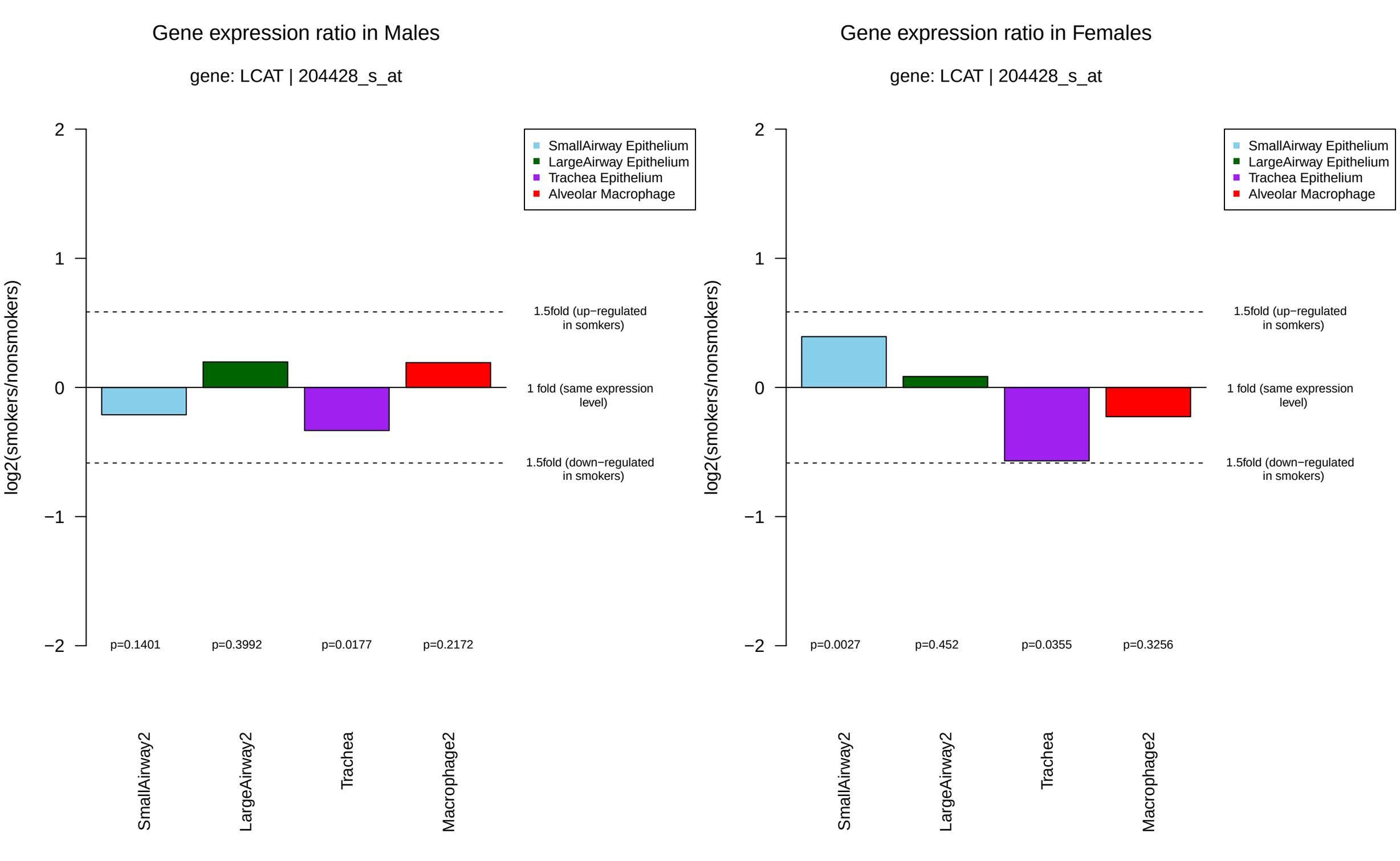
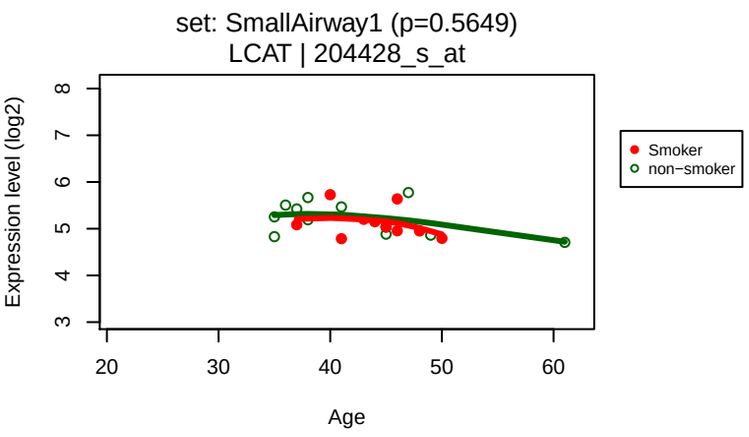
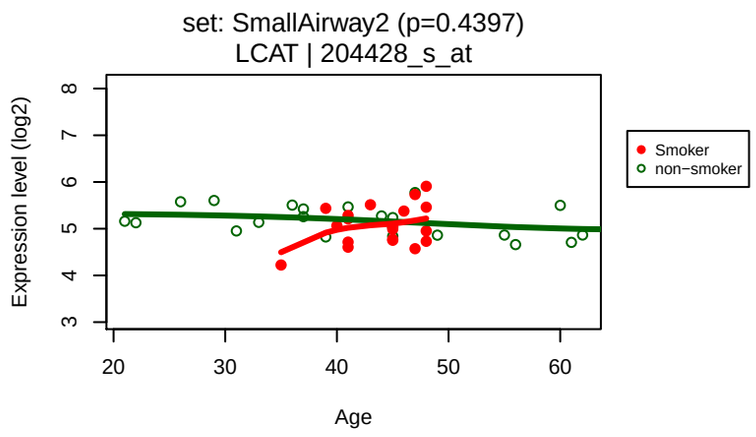
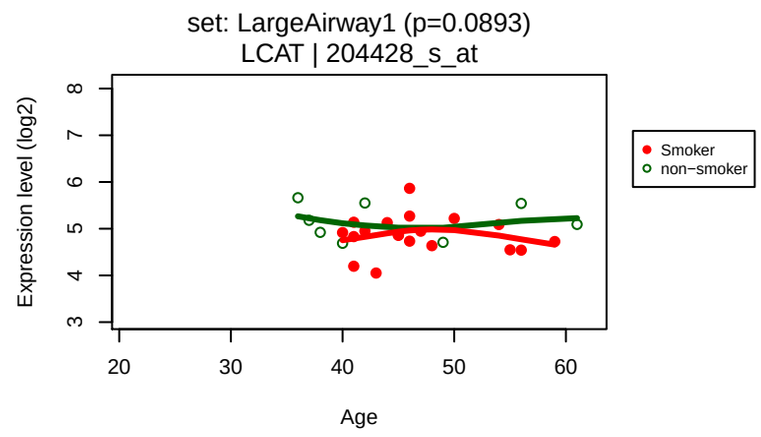

Gene expression of smokers and nonsmokers
gene: LCAT | 204428_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5649
p=0.4397
p=0.0568
p=0.0893
p=0.3005
p=0.001
p=0.412
p=0.6582
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

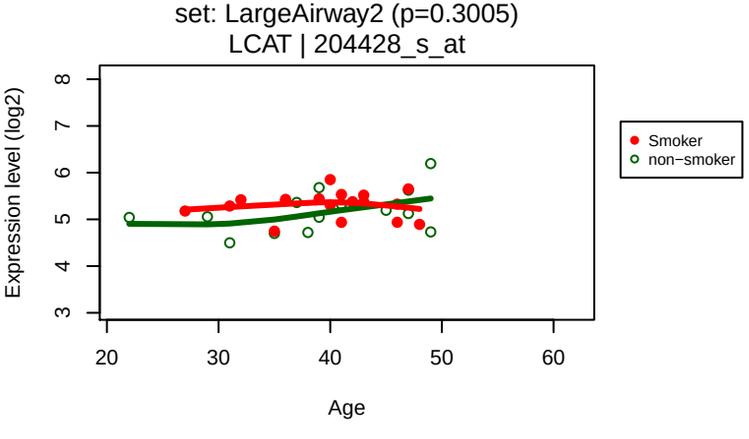

Gene expression ratio (smokers/nonsmokers)
gene: LCAT | 204428_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5649
p=0.4397
p=0.0568
p=0.0893
p=0.3005
p=0.001
p=0.412
p=0.6582
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

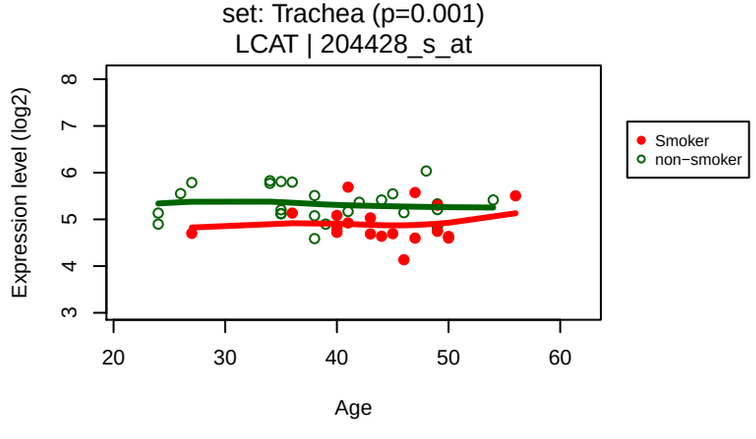

Gene expression ratio in Males
gene: LCAT | 204428_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1401
p=0.3992
p=0.0177
p=0.2172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LCAT | 204428_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0027
p=0.452
p=0.0355
p=0.3256
SmallAirway2
LargeAirway2
Trachea
Macrophage2

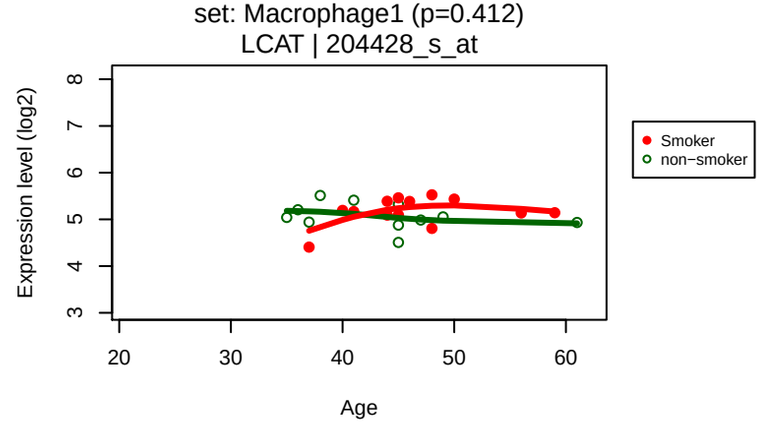

set: SmallAirway1 (p=0.5649)
LCAT | 204428_s_at
Expression level (log2)
Age
Smoker
non−smoker

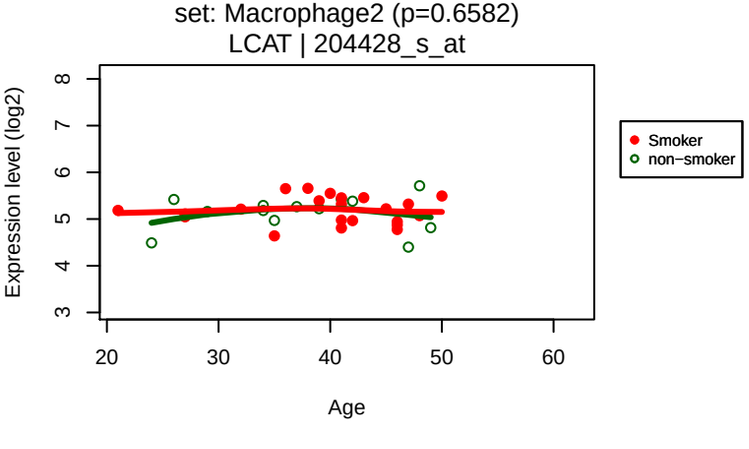

set: SmallAirway2 (p=0.4397)
LCAT | 204428_s_at
Expression level (log2)
Age
Smoker
non−smoker

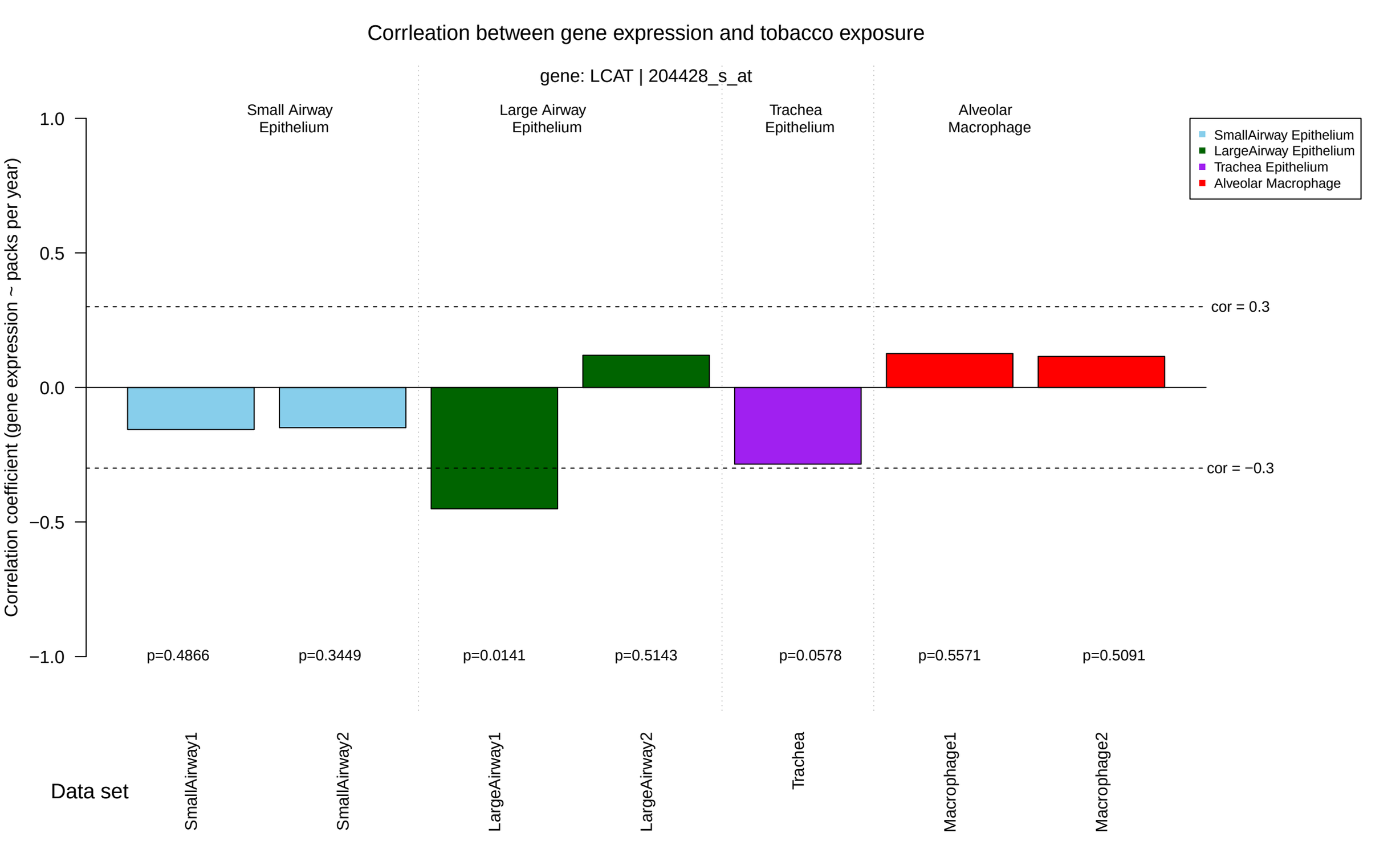

set: LargeAirway1 (p=0.0893)
LCAT | 204428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3005)
LCAT | 204428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.001)
LCAT | 204428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.412)
LCAT | 204428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6582)
LCAT | 204428_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LCAT | 204428_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4866
p=0.3449
p=0.0141
p=0.5143
p=0.0578
p=0.5571
p=0.5091
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2